Memory and Forgetting



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AHUL261

Memory

- Without memory….
  - No recollection of events
  - No knowledge
  - No reflection of past events; no giving advice to others
  - No basis for self-esteem and mood
  - No idea about past achivements

At the end of this Chapter you should be able to:

- Understand what memory is
- Learn about working memory and long term memory
- Learn about three aspects of memory; encoding, storage and retrieval
- Understand what happens when memory fails

Memory

- What is memory?
- Are there different kinds of memory?
- What are contemporary approaches to memory?

Memory

- The process by which we encode, store, and retrieve information.

The Memory Process

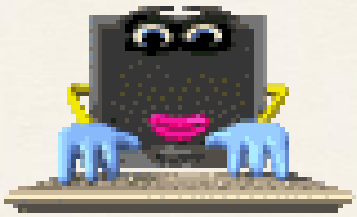
Three step process....

1. **Encoding**: The processing of information into the memory system.

2. **Storage**: The retention of encoded material over time.

3. **Retrieval**: The process of getting the information out of memory storage.







Three System approach to Memory (Atkinson & Shiffrin Model)

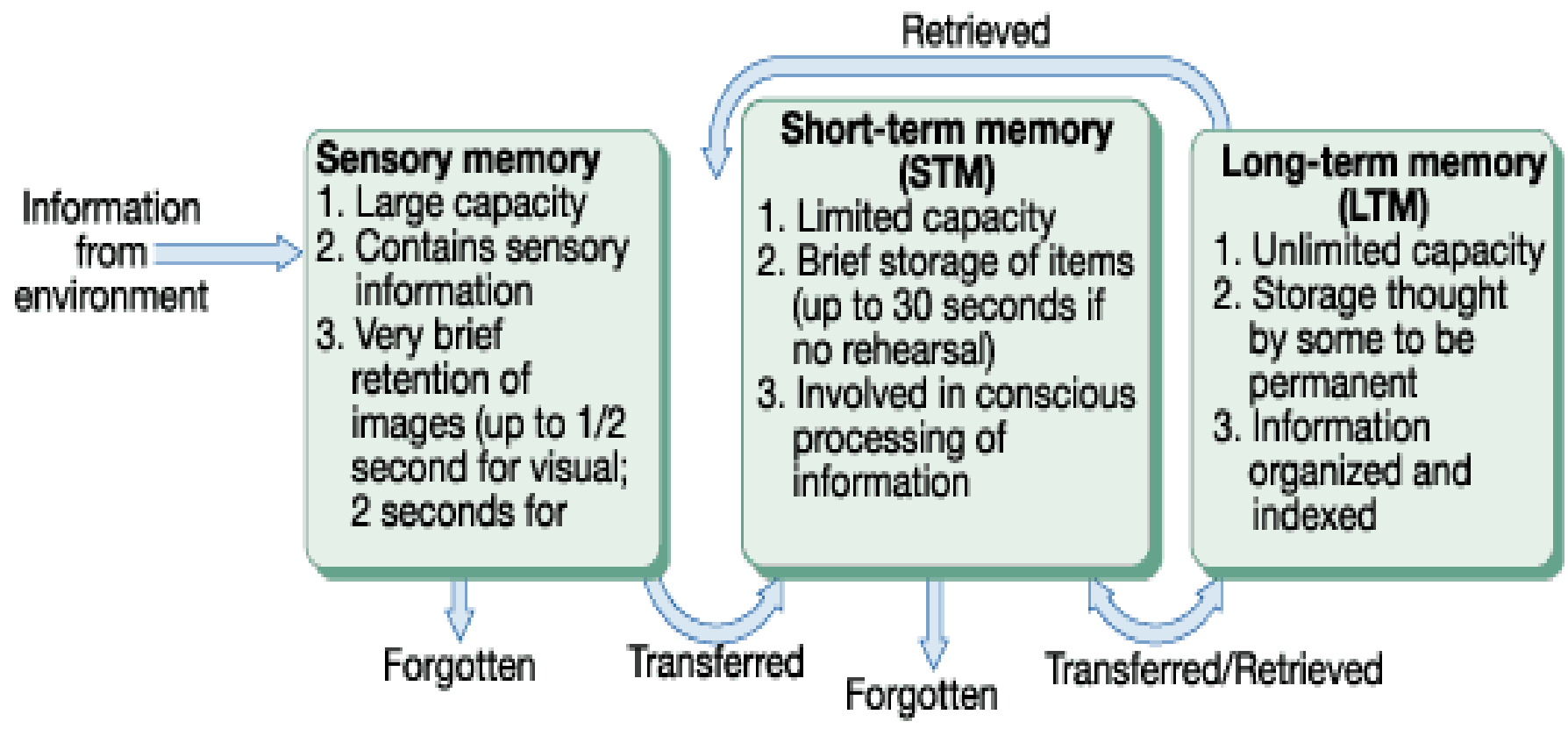




FIGURE 1 Memory is built on three basic processes—encoding, storage, and retrieval—that are analogous to a computer's keyboard, hard drive, and software to access the information for display on the screen. The analogy is not perfect, however, because human memory is less precise than a computer. How might you modify the analogy to make it more accurate?

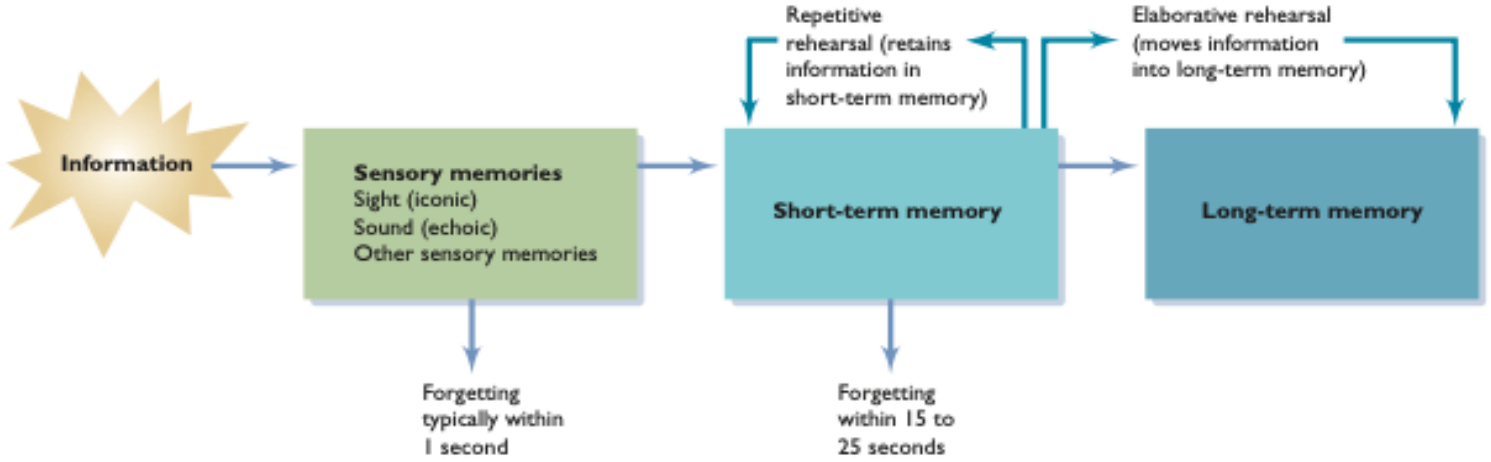


FIGURE 2 In this three-stage model of memory, information initially recorded by the person's sensory system enters sensory memory, which momentarily holds the information. The information then moves to short-term memory, which stores it for 15 to 25 seconds. Finally, the information can move into long-term memory, which is relatively permanent. Whether the information moves from short-term to long-term memory depends on the kind and amount of rehearsal of the material that is carried out. (Source: Atkinson & Shifrin, 1968.)

Sensory Memory

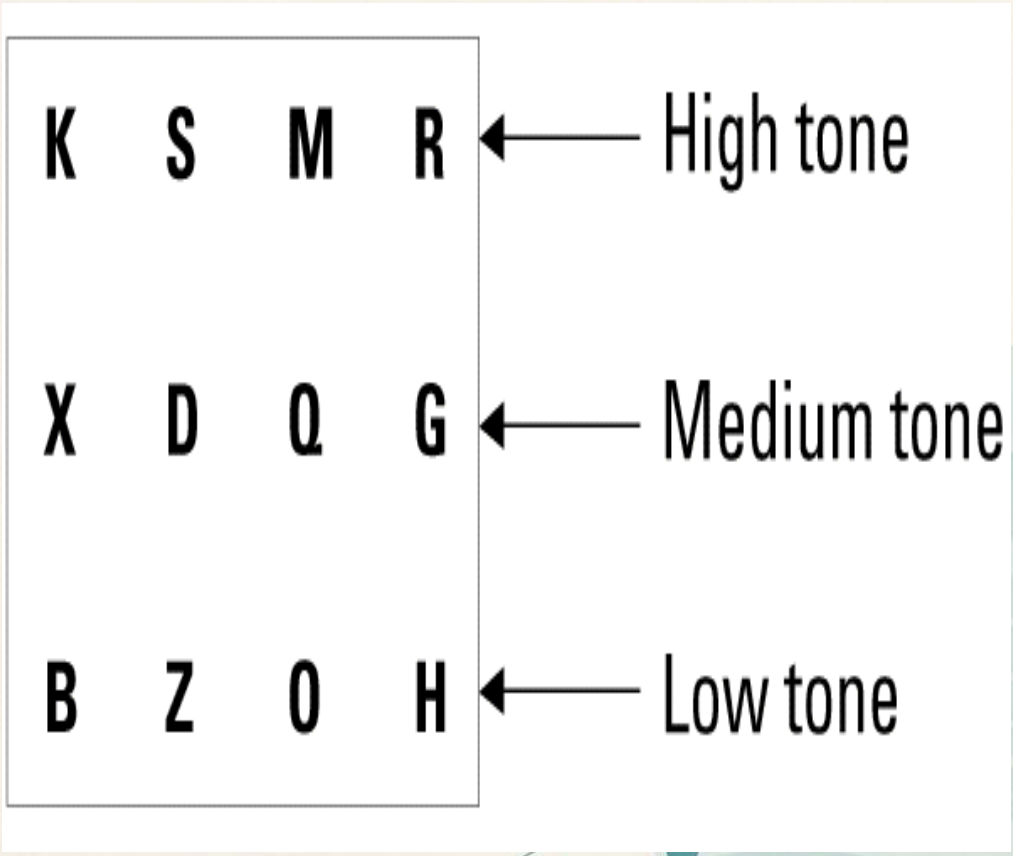
- **Sensory Memory**
  - Records information from the senses for up to three seconds
  -
  - **Examples are Iconic (Visual) Memory and Echoic (Auditory) Memory**



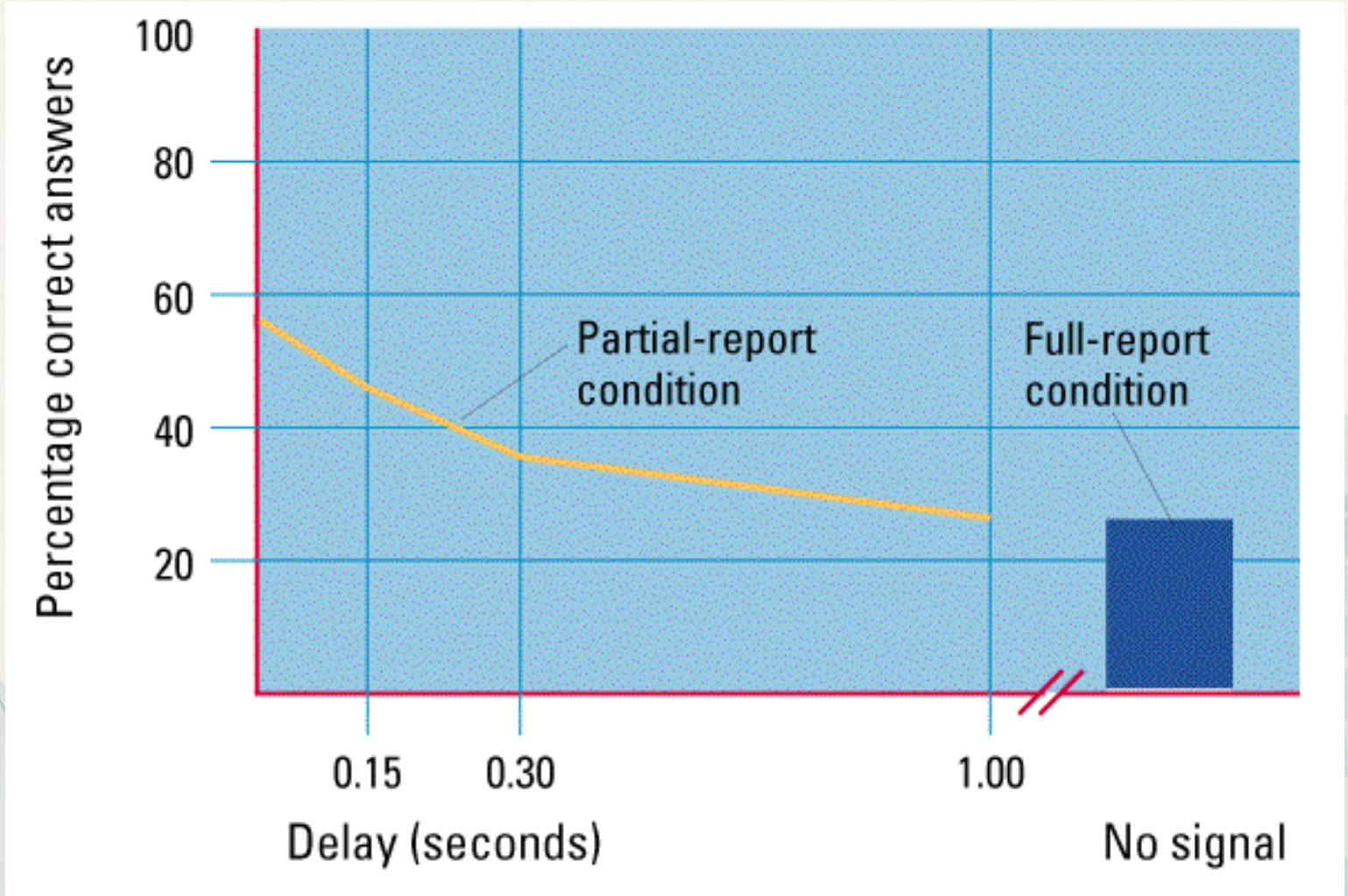
A momentary flash of lightning leaves a sensory visual memory, a fleeting but exact replica of the stimulus that fades away

The Sensory Register
Testing for Iconic Memory

- Invented by George Sperling
- A letter array is shown briefly
- After array is gone, tone signals which row to report
- Subjects recalled more letters when signaled to recall only one row compared to trying to recall all the letters



The Sensory Register Duration of Iconic Memory



Short Term Memory



- The stuff we encode from the sensory goes to STM.
- Events are encoded visually, acoustically or semantically.
- Holds about 7 (plus or minus 2) items for about 20 seconds.
- We recall digits better than letters.

Working Memory

- When we are actively working, we want information to be immediately available to us. In our brain, this happens in Working (short-term) Memory.

- Long Term Memory is for extra information. It contains everything you know. It is storage for information that is not used right now but may be needed later.

Storage Capacity of Working and Long-term memory

- Long-term memory capacity: huge
- Working memory capacity: more modest
- Memory span: way of measuring working memory capacity
  - Random, unrelated information: we can store about 7, plus or minus 2, items (5 – 9 items)'
  - Referred to as "the magic number 7"

Ways to remember things in STM…so they go to LTM

- **Chunking**: Organizing items into familiar, manageable units.

1-4-9-2-1-7-7-6-1-8-1-2-1-9-4-1

- **Mnemonic devices**

VIBGYOR

- **Rehearsal**

The repetition of information that has entered short term memory

Chunking

- Working memory can only handle a small number of items at one time. However what these packages contain can be up to us.

9182736455463728190

Can you remember all of these digits?

Chunking

- If you look at these digits independently, you will fail to remember them!
- What about if you organize them differently?

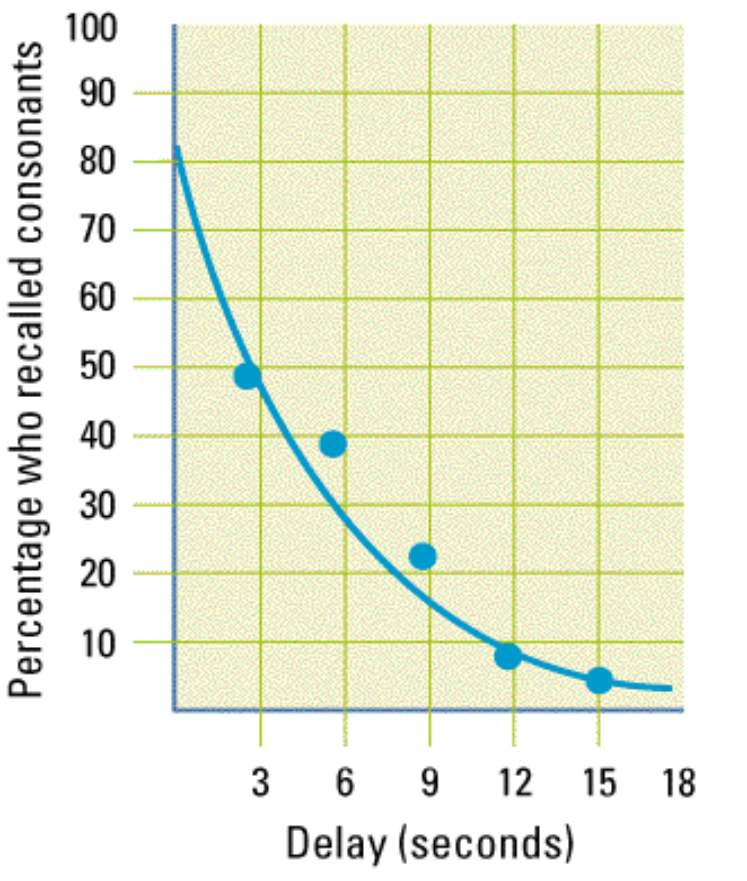
9 18 27 36 45 54 63 72 81 90

Now you only have to remember the relationship between these items. So working memory's capacity is dependent on these chunks, not independent items.

Short-Term Memory

Duration of Short-Term Memory

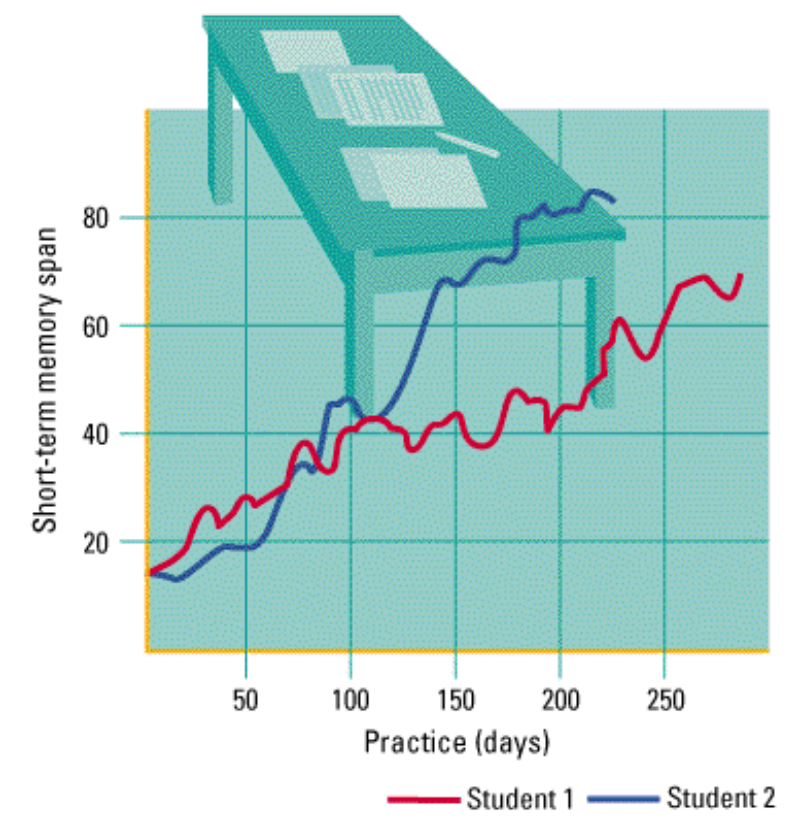
- Subjects memorized nonsense syllables, (e.g., MJK, ZRW).
- To prevent rehearsal, they were given a distracter task during the waiting period.
- When a cue was given, subjects tried to recall the letters.
- Short-term memories vanish within twenty seconds.



Short-Term Memory

Capacity Increased Memory Span

- Two students practiced memory span tasks for an hour 3-4 days/week.
- After six months, digit span had increased from 7 to 80 items.

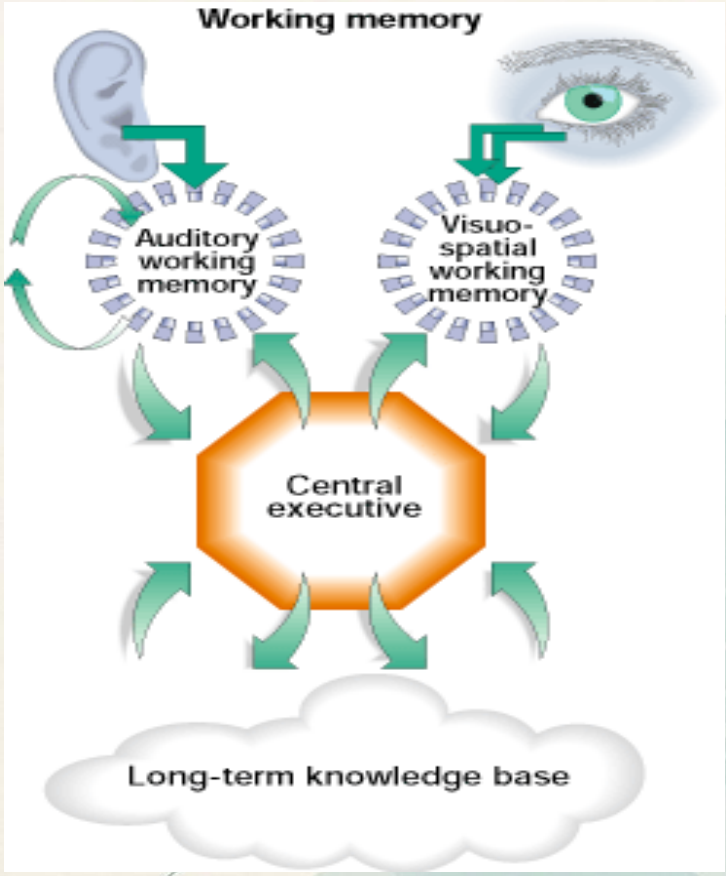


Short-Term Memory

Functions of Short-Term Memory

Working memory -Term used to describe short-term memory as an active workspace where information is accessible for current use.

- Baddeley's model of working memory contains three elements:
  - **A "central executive"**
  - **Auditory working memory**
  - **Visuo-spatial working memory**
- Material can enter conscious workspace from senses or from long-term memory



Working Memory

- “Loading platform” analogy
- Long term memory must be “loaded” or “pass through” WM
- How does it move? How is it transformed into Long term memory?
  - Rehearsed
  - Chunked
- Primacy Effect – items presented early are remembered better
- Recency Effect - items presented last are remembered better

Evidence for existence of Short-Term Memory

The Serial-Position Effect

- **Serial Position Curve**
  - Indicates the tendency to recall more items from the beginning and end of a list than from the middle.

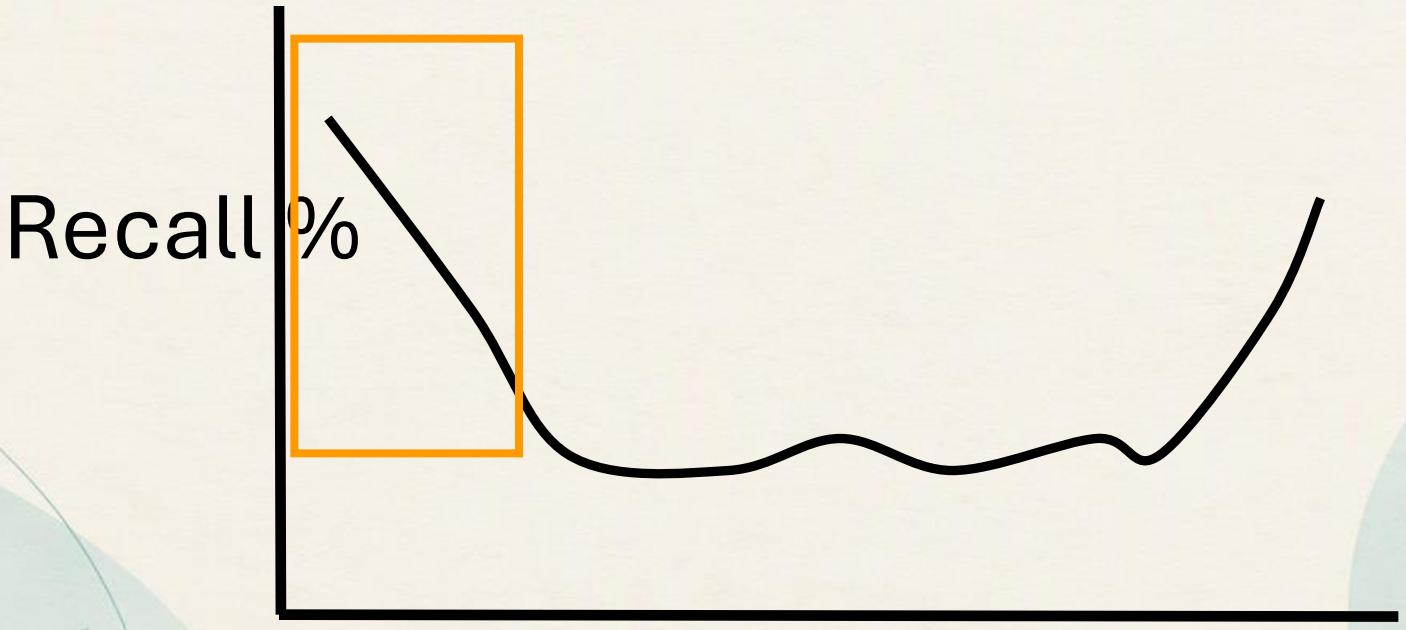
- Both groups of subjects showed primacy effects, good recall of first items on list.

- Only the no-delay group showed recency effects, good recall for last items.



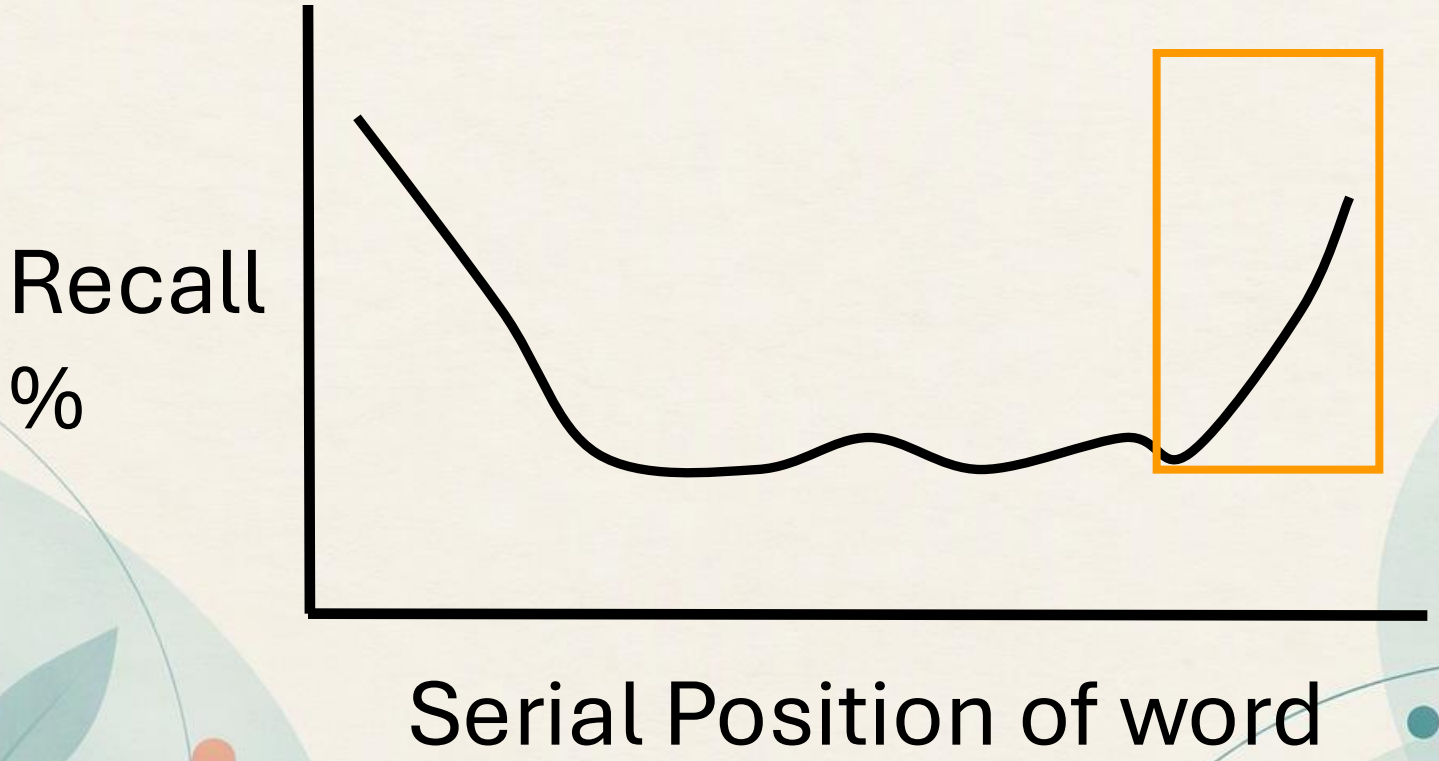
Serial Position

Primacy Effect



Serial Position

Recency Effect



Long Term Memory



- Unlimited storehouse of information.
- Explicit (declarative) memories
- Implicit (non-declarative) memories

Levels of processing theory

- An alternative to structural models of memory, focussing instead on memory processes.
- Levels of processing theory suggests that information can be processed at different levels, and that the way in which memory is processed can affect the likelihood of it being retrieved in the future (Craik and Lockhart,1972).

Levels of processing theory

- Deep processing:
  - Meaning-based attention
  - Anything that connects new information to already-learned material
  - Material that “makes sense” will be encoded more efficiently
  - Results in superior recall

Levels of processing

- Memory connections:
  - Links among ideas
  - Abstract similarities
- When the time comes to recall something, these connections, established during initial learning or acquisition, can be used as retrieval paths.
- Mnemonics
  - “Method of loci”
  - Based on rhythm/rhyme/melody/visualization



A

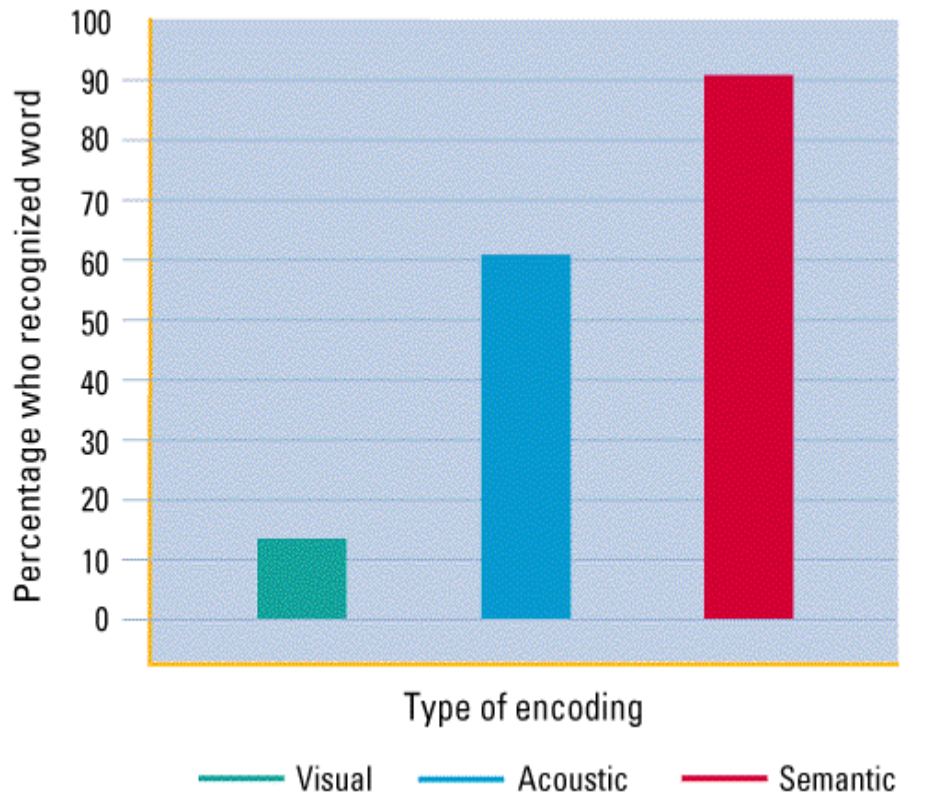


B

Research participants shown related elements, such as a doll sitting on a chair and waving flag (A), are more likely to associate the words doll, flag and chair than participants who are shown the three objects next to each other but not interacting (B).

Long-Term Memory
Encoding
Elaborative Rehearsal

- Subjects were shown lists of words and asked to use one of three strategies:
  - **Visual: Is the word printed in capital letters?**
  - **Acoustic: Does the word rhyme with _____?**
  - **Semantic: Does the word fit the sentence _____?**
- The more thought involved (elaborative rehearsal), the better was their memory.



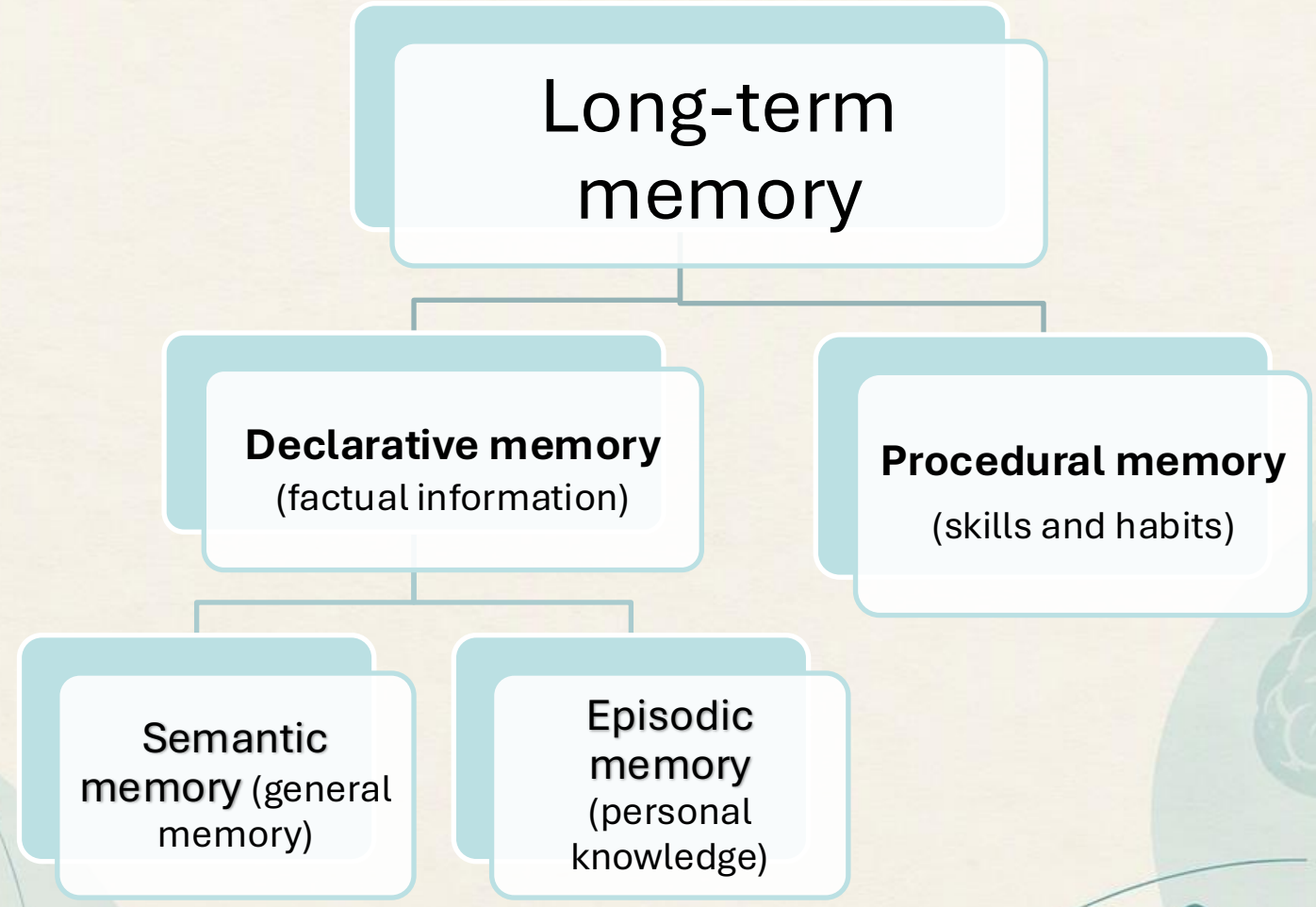
Long-Term Memory

Storage

- **Procedural Memory**
  - **Stored long-term knowledge of learned habits and skills.**
  - Examples are how to drive, ride a bike, tie one's shoes, etc.
- **Declarative Memory**
  - **Stored long-term knowledge of facts about ourselves and the world.**
  - Includes both semantic (nonpersonal) and episodic (personal) memories

Long-Term Memory

Types

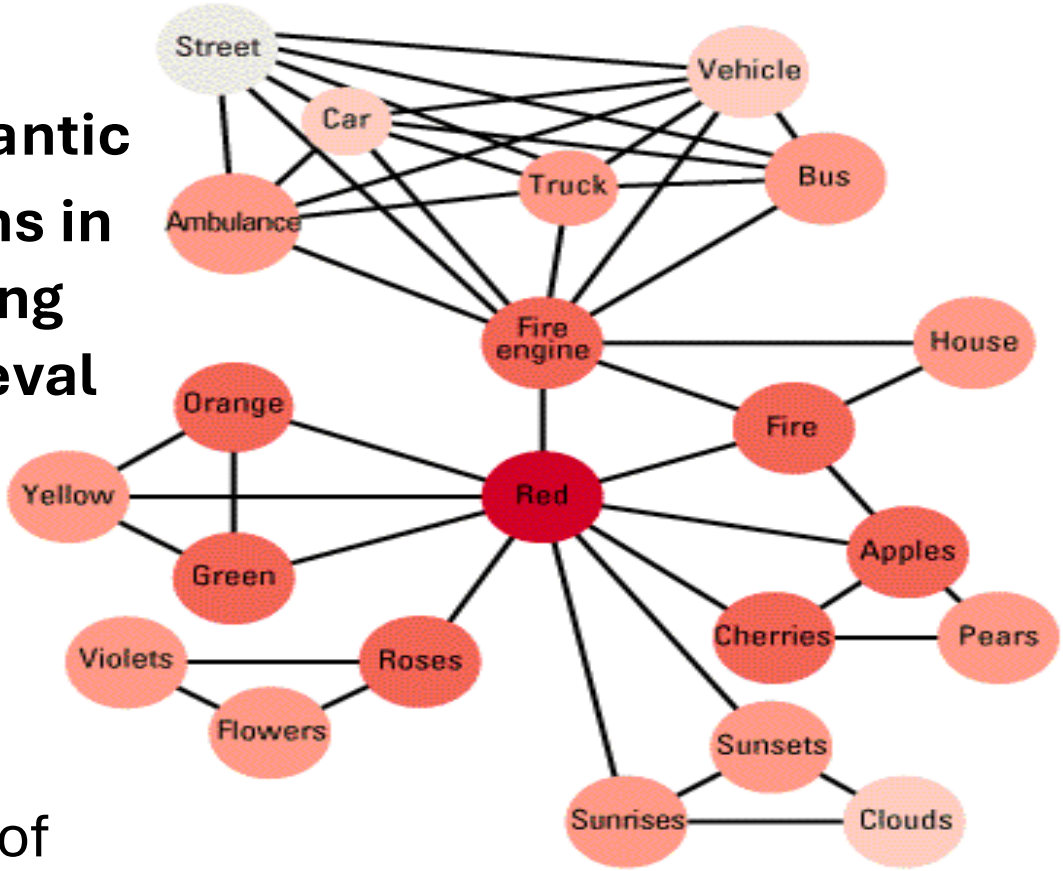


Long-Term Memory

Storage

Semantic Networks

A complex web of semantic associations that link items in memory such that retrieving one item triggers the retrieval of others as well



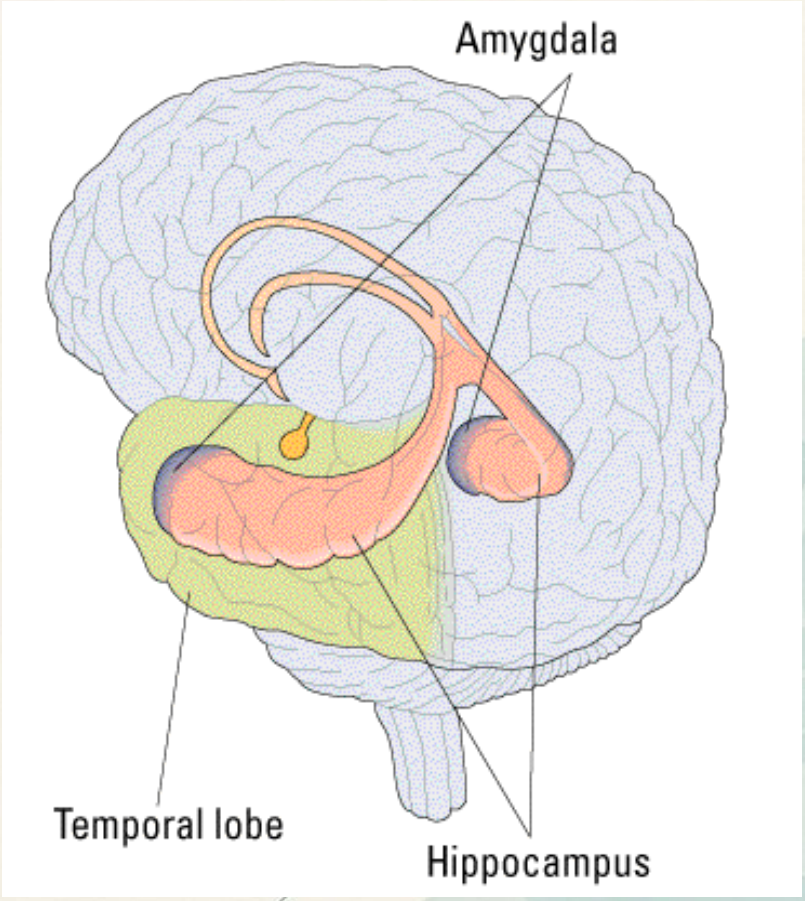
Mental representations of inter-connected information

Long-Term Memory

Storage

The Hippocampal Region

- **Hippocampus:** Part of the limbic system that plays a key role in encoding and transferring new information into long-term memory.
- **Anterograde amnesia**
  - Inability to store new information
- **Retrograde amnesia**
  - Inability to retrieve memories from the past



Recalling long-term memories

Retrieval Cues

- Recall
  - A specific piece of information must be retrieved
- Recognition
  - Occurs when one is presented with a stimulus and asked whether he has been exposed to it previously or is asked to identify it from a list of alternatives

Retrieval cues

Retrieval failure:

One type: the "**tip of the tongue**" phenomena

- What was the name of the character played by Amitabh Bacchan in Sholay?

- Who was the name of Spiderman aka Pater Parker's uncle?

Long-Term Memory
Retrieval

- **Explicit Memory**
  - **The types of memory elicited through the conscious retrieval of recollections in response to direct questions.**
  - Conscious retention, direct tests, disrupted by amnesia, encoded in the hippocampus
- **Implicit Memory**
  - **A nonconscious recollection of a prior experience that is revealed indirectly, by its effects on performance.**
  - Nonconscious retention, indirect tests, intact with amnesia, encoded elsewhere

Explicit Memories

- Episodic Memories



- Semantic Memories





Implicit Memories



- Procedural Memories

- Conditioned Memories





Long-Term Memory

Retrieval

Context-Dependent Memory

- Russian-English bilinguals were prompted in English and in Russian to recall stories.

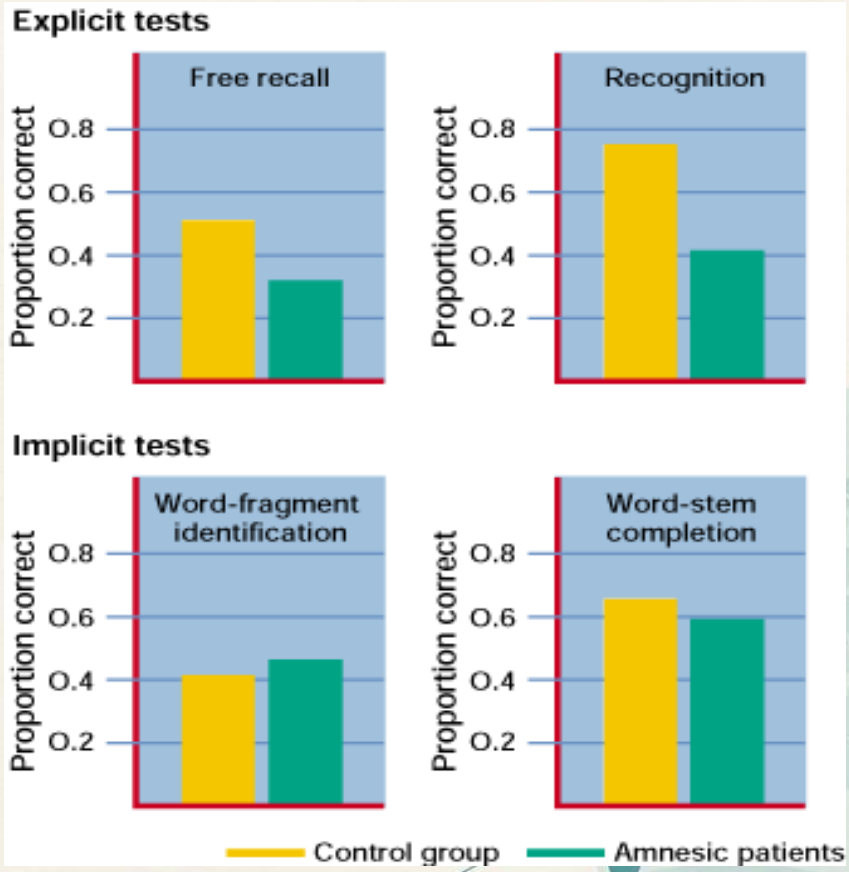
- They recalled more Russian-experienced events when interviewed in Russian and more English-experienced events when interviewed in English.

Long-Term Memory

Retrieval

Retention Without Awareness

- Amnesic patients and normal controls were tested for memory of words learned previously.
- Amnesics performed poorly on explicit memory tasks.
- However, performance on implicit memory tasks was similar to control subjects.



The Context Matters!!!





- Flashbulb Memories
- Mood Congruent Memory
- State Dependent Memory









“Flashbulb memories”

- Flashbulb memories are related to special events, usually unexpected and emotionally strong. Such as 9/11 terrorist attacks to the World Trade Center, 2008 attack on Taj Mahal hotel in Mumbai, etc.
- Where and who were you with when you first heard these events? Most people remember all the details…
- *Especially* vivid memories
- Focus: immediate and personal details

Long-Term Memory

Retrieval

Implicit Memory in Everyday Life

- Déjà vu
  - A sense of familiarity but no real memory
- The false-fame effect
  - Names presented only once, familiarity but no real memory, assume person is famous
- Eyewitness transference
  - Face is familiar, but situation in which they remembering seeing face is incorrect
- Unintentional plagiarism
  - Take credit for someone else's ideas without awareness

Long-Term Memory

Reconstruction

“Office” Schema

- Study this picture for 30 seconds.



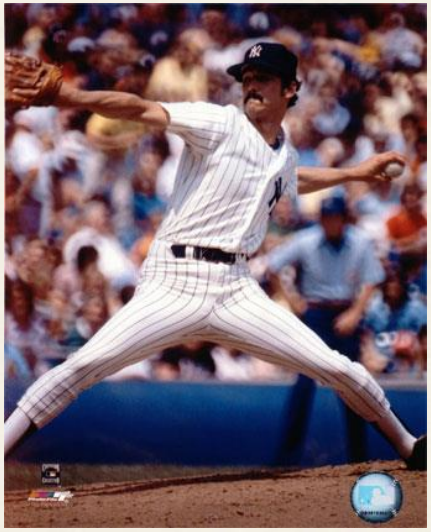
List as many objects as you can recall from the photograph you just saw.

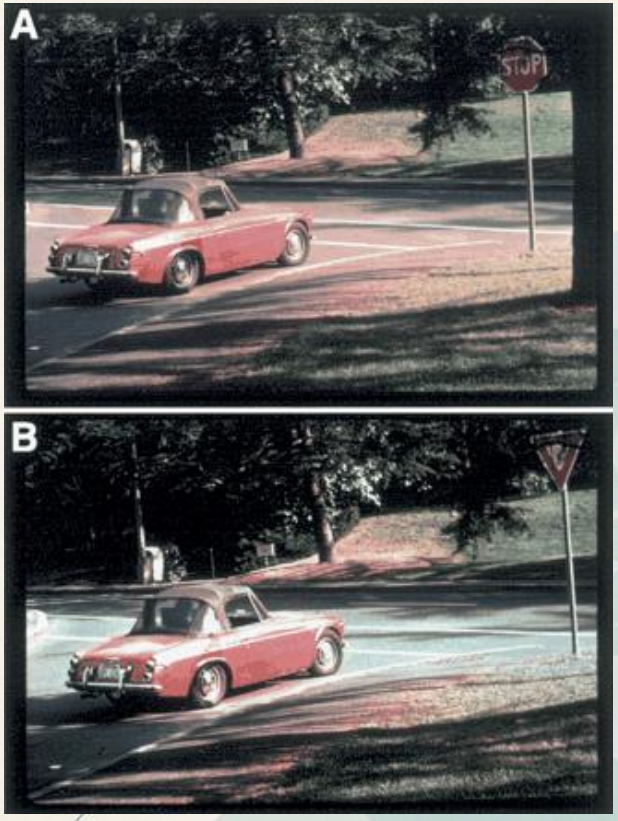
Constructive Processes in Memory



- Memories are not always what they seem.
- Elizabeth Loftus
- A constructed memory is a created memory.
- Misinformation effect







Misinformation Effect

- It is common to ‘remember’ things that never happened.
- Example of this is experiment by: Wade, Garry, Read, & Lindsay (2002)
- People were shown photographs of themselves in a hot-air balloon and asked what they remembered. Most people remembered the experience. But the photos were made using photoshop. The people had never been in a hot-air balloon

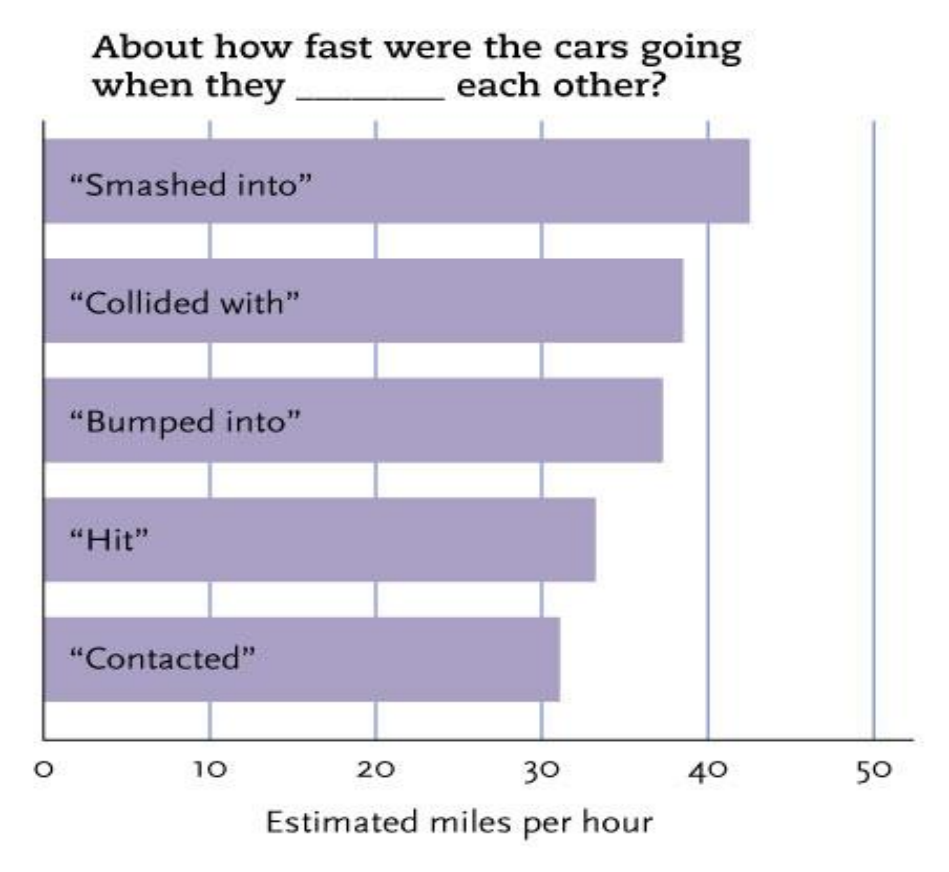


Memory in the Courtroom: The Eyewitness on Trial

- Mistaken identity
- Impact of Weapons
- Specific wording of questions
  - Children witnesses

Memory in the Courtroom: The Eyewitness on Trial

Accuracy of Eyewitness Testimony Affected by Questioner's Word Choice



Repressed and False Memories: Separating Truth from Fiction

- Repressed and False Memories
  - Recollections of events that are initially so shocking that the mind responds by pushing them into the unconscious
  - Memories may be inaccurate or even wholly false.
  - Controversy regarding their legitimacy

Autobiographical Memory: Where Past Meets Present

- Recollection of circumstances and episodes from our own lives
  - One tends to forget information about one's past that is incompatible with the way in which one currently sees oneself.





Forgetting: When Memory Fails

- Why do we forget information?
- What are the major memory impairments?

Why we forget

- **Forgetting**: we knew it once, but no longer

- Inadequate encoding: Forgetting can often be traced to poor or missing strategies for encoding

- Cue-dependent Forgetting
  - Occurs when there are insufficient retrieval cues to rekindle information that is in memory

Why we forget

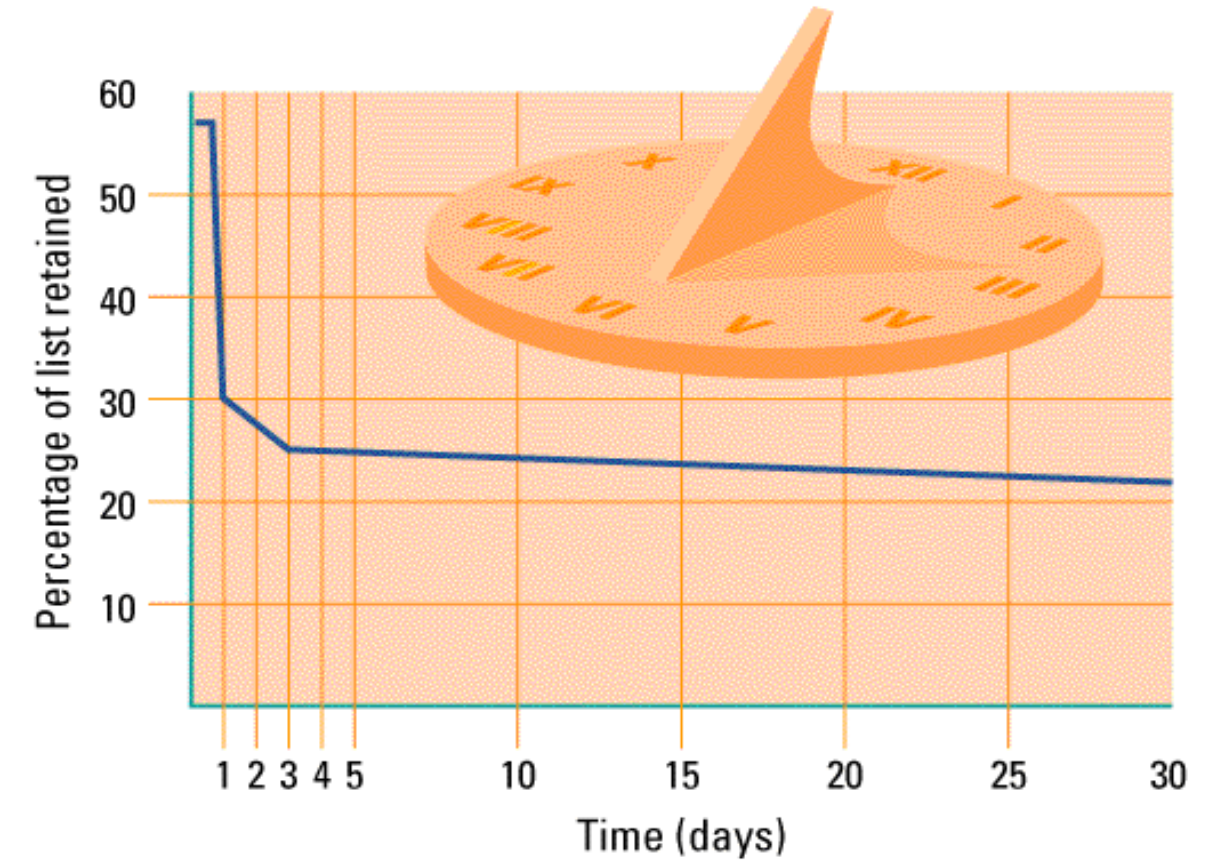
- **Decay:** a process that occurs on a cellular level by normal metabolic "wear and tear" on cells involved with memory

- **Interference:** New learning interferes – independent of the passage of time
  - Passage of time not a powerful factor in explaining forgetting
  - Number of intervening events are more useful variable to examine to explain forgetting

Long-Term Memory

Forgetting

The Ebbinghaus Forgetting Curve

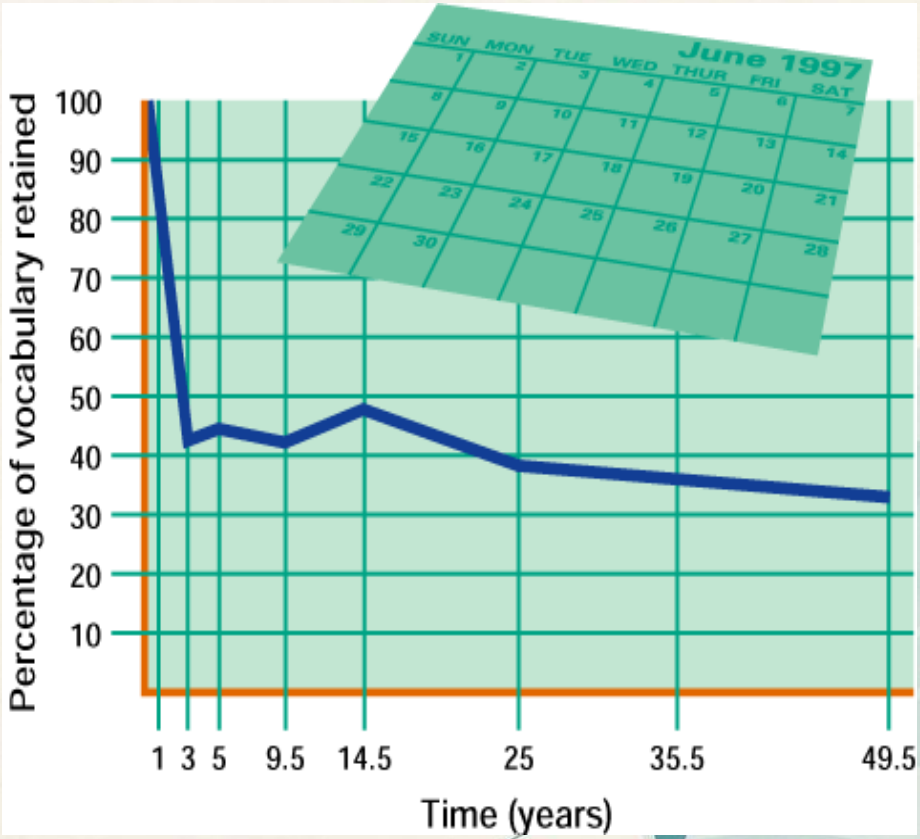


Long-Term Memory

Forgetting

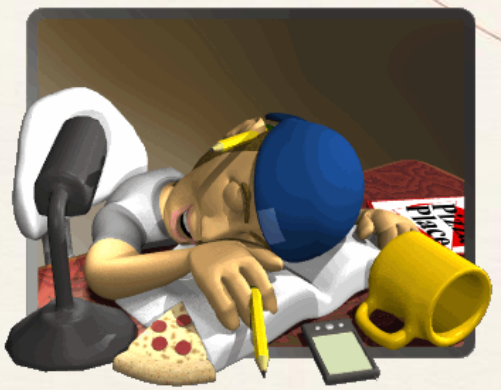
Long-Term Forgetting Curve

- How much Spanish vocabulary is remembered over time?
- Most forgetting occurs within the first three years.
- After that, memory remains stable.

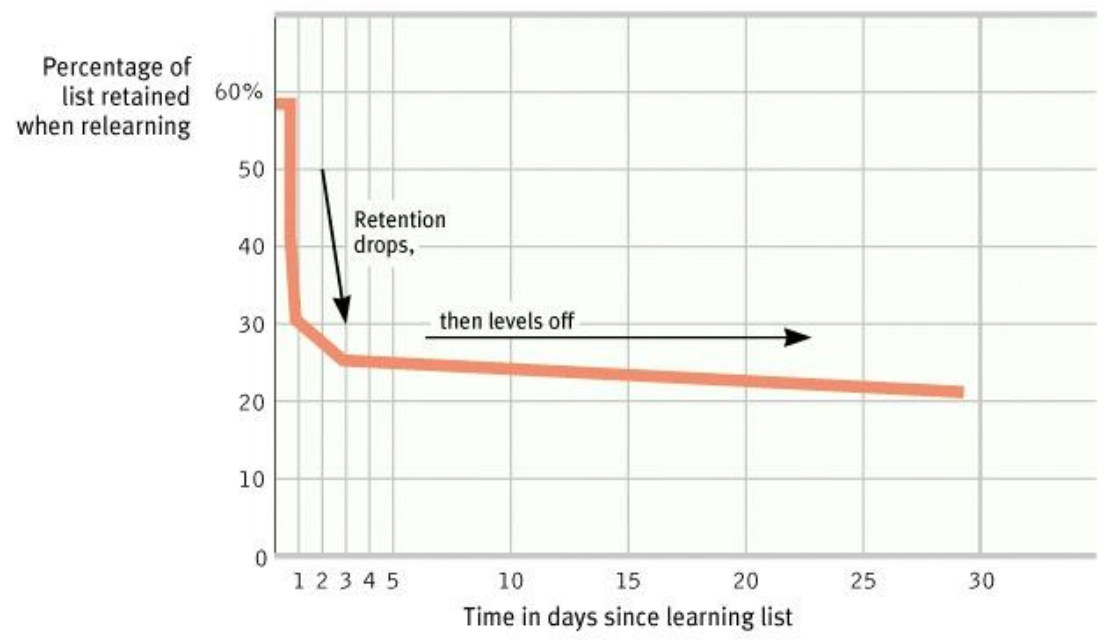


Spacing Effect

- DO NOT CRAM!!!!!!!!!!!!



- **Ebbinghaus's Forgetting Curve**



Improving Memory

- Do not believe claims about drugs that improve memory.

- Effective strategies
  - Organization cues
  - Keywords
  - Elaborative rehearsal
  - Effective note taking

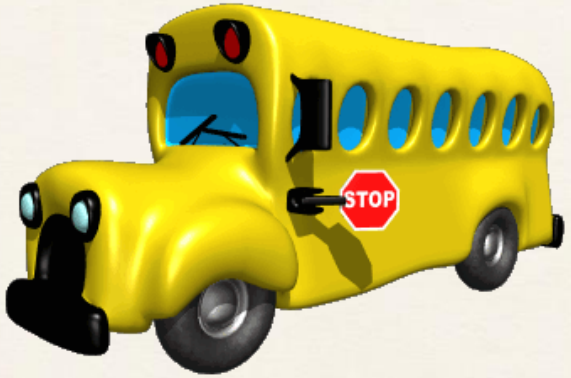
Proactive and Retroactive Interference: The Before and After of Forgetting

- Retroactive Interference
  - Difficulty in the recall of information because of later exposure to different material

- Proactive Interference
  - Information learned earlier disrupts the recall of newer material.

Forgetting

Getting a new bus number and forgetting old bus number.



- **Retroactive Interference**: new information blocks out old information.

- **Proactive Interference**: old information blocks out new information.



Calling your new girlfriend by old girlfriends name.

Memory Dysfunctions: Afflictions of Forgetting

- Alzheimer’s Disease
- Amnesia
  - Retrograde
  - Anterograde

Memory Dysfunctions: Afflictions of Forgetting

- Alzheimer's Disease

The causes of Alzheimer's disease are not fully understood. Increasing evidence suggests that Alzheimer's results from an inherited susceptibility to a defect in the production of the protein beta amyloid, which is necessary for the maintenance of nerve cell connections. When the synthesis of beta amyloid goes awry, large clumps of cells form, triggering inflammation and the deterioration of nerve cells in the brain

Amnesia : Memory Loss

- Different brain tissue supports implicit memories as compared to explicit memories
- Evident when studying anterograde amnesia
  - Lesions in hippocampus and temporal cortex: create anterograde amnesia
  - Lesions from other types of brain injury: create retrograde amnesia

  Supports the theory that different brain structures/regions “handle” different types of memory

Retrograde amnesia



- Forgetting the past, but being able to make new memories. Soldiers are sometimes unable to remember their experiences in battle-even the ones that occured a day before.

Anterograde Amnesia

- Anterograde means ‘in a forward direction’. May be caused by a stroke or a physical trauma.
- It is essentially an inability to learn anything new/make new memories.

Summary

- We learnt that memory refers to the processes of encoding, storage, and retrieval
- We learnt about theories that explain the memory processes.
- We also encountered several phenomena relating to memory, including the tip-of-the-tongue phenomenon and fl ashbulb memories.
- Above all, we observed that memory is a constructive process by which interpretations, expectations, and guesses contribute to the nature of our memories.